Computational Photography 2026

Tutorial 2

You should choose **one task** from the 3 tasks below, you will find the corresponding inputs for each task in the attachment. For algorithm designs, you can go to Google Scholar referring to corresponding papers, but you must implement the algorithm **in Python by yourself**. Please **write a report** to give a detailed introduction of your task and solution to the problem. No plagiarism is allowed.

1. High-dynamic-range (HDR) Imaging

Multiple standard-exposure images

HDR image

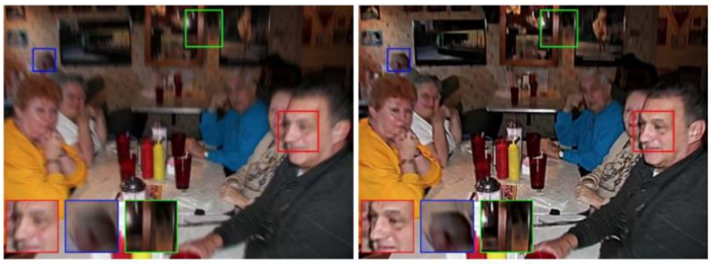
Dynamic range, abbreviated DR, is the ratio between the largest and smallest values that a certain quantity can assume. Apparently, revealing detail in the darkest of shadows requires high exposures, while preserving detail in very bright situations requires very low exposures. Most cameras cannot provide this range of exposure values within a single exposure, thus their DRs are relatively low.

In most imaging devices, the degree of exposure to light applied to the active element (be it film or CCD) can be altered in one of two ways: by either increasing/decreasing the size of the aperture or by increasing/decreasing the time of each exposure. Exposure variation in an HDR set is only done by altering the exposure time and not the aperture size; this is because altering the aperture size also affects the depth of field. Hence, high-dynamic-range photographs are generally achieved by capturing multiple standard-exposure images and then later merging them into a single image.

Please write code to merge the given photos with different exposure times into a HDR image. You can compare your own result with an HDR camera if possible.

2. Camera Shake Deblurring

Camera shake occurs when a camera is unintentionally moved while the shutter speed of the camera is too low to capture a perfectly still image. The resulting image either looks blurry, or shows traces of ”dragging” of the subjects, where multiple outlines could be visible around the subjects. Camera shake due to slow shutter speed typically takes place in low-light conditions when there is not enough light for the camera to yield a sufficiently bright image, or when exposure settings are set up incorrectly. For example, it is often needed to open up lens aperture as much as possible and increase camera ISO when shooting in indoor environments, because the amount of ambient light is too low.



Blurry image Deblurring result

Mathematically, the motion blur caused by camera shake is typically modeled as the convolution of a (sometimes space- or time-varying) point spread function with a hypothetical sharp input image, where both the sharp input image (which is to be recovered) and the point spread function are unknown. In almost all cases, there is insufficient information in the blurred image to uniquely determine a plausible original image, making it an ill-posed problem. In addition the blurred image contains additional noise which complicates the task of determining the original image. This is generally solved by the use of a regularization term to attempt to eliminate implausible solutions.

Please write code to realize motion deblurring for the given photos. Try your best to get better results.

3. Low Illumination Image Enhancement

Low illumination image enhancement is highly desired for computer vision applications. For example, due to the intrinsic weak ambient light, images captured at night have low brightness, low contrast and high noise. One way to cope with this problem is using artificial lighting or a long exposure, while low illumination image can also be enhanced by proposing algorithms so that the brightness and contrast of images can be enhanced significantly and noise is reduced effectively.

Low illumination image enhancement

Please write code to realize image enhancement on the given photos. Try your best to get better results.